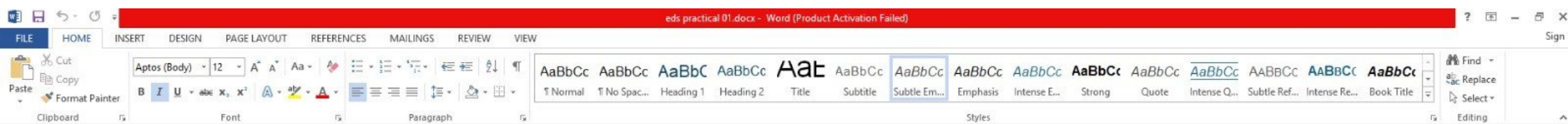


```
/
1) mass=float(input("Enter mass in kilogram:"))
   Enter mass in kilogram:10
   velocity=float(input("Enter velocity in meter per second:"))
   momentum=mass*velocity
   print(f"momentum of the object is:{momentum}")
```

```
1) import math
n=int(input("enter a number:"))
if 0<=n<10:
    print(f"square of {n}:{n**2}")
elif 10<=n<100:
    print(f"square root of {n}:{math.sqrt(n):.2f}")
elif 100<=n<1000:
    print(f"cube root of {n}:{n**(1/3):.2f}")
else:
    print("please enter a number between 0 to 999.")
```

```
num=[10,20,30,40]
print("list",num)
list [10, 20, 30, 40]
num.append(60)
num=[10,20,30,40]
print("list",num)
list [10, 20, 30, 40]
num.append(60)
print("after append",num)
after append [10, 20, 30, 40, 60]
num.insert(3,35)
print("after insert",num)
after insert [10, 20, 30, 35, 40, 60]
num.pop()
60
print("after pop",num)
after pop [10, 20, 30, 35, 40]
num.reverse()
print("after reverse",num)
after reverse [40, 35, 30, 20, 10]
num.reverse(20)
num.remove(20)
print("after remove",num)
after remove [40, 35, 30, 10]
s=num[1:2]
print("slice:",s)
|
```



```
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```
Python 3.11.2 (tags/v3.11.2:878ead1, Feb 7 2023, 16:38:35) [MSC v.1934 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>> from datetime import date
>>>
>>> def calculate_age(birth_year):
...     current_year=date.today().year
...     age=current_year-birth_year
...     return age
...
>>> def convert_to_dollar(rupees):
...     dollars=rupees/83
...     return dollars
...
>>> birth_year=int(input("enter birth year:"))
enter birth year:
Traceback (most recent call last):
  File "<pyshell#11>", line 1, in <module>
    birth_year=int(input("enter birth year:"))
ValueError: invalid literal for int() with base 10: ''
>>> salary_rupees=float(input("enter salary in rupees:"))
enter salary in rupees:
Traceback (most recent call last):
  File "<pyshell#12>", line 1, in <module>
    salary_rupees=float(input("enter salary in rupees:"))
ValueError: could not convert string to float: ''
>>>
>>> age=calculate_age(birth_year)
Traceback (most recent call last):
  File "<pyshell#14>", line 1, in <module>
    age=calculate_age(birth_year)
NameError: name 'birth_year' is not defined
>>> |
```

```
1) mass=float(input("Enter mass in kilogram:"))
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```
Enter mass in kilogram:10
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```
velocity=float(input("Enter velocity in meter per second:"))
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momentum=mass*velocity
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```
print(f"momentum of the object is:",{momentum})
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    print(f"square root of (n):{math.sqrt(n):.2f}")
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    print(f"cube root of {n}:{n**(1/3):.2f}")
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else:
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    print("please enter a number between 0 to 999.")
```

```
3 from datetime import date
```

```
def calculate_age(birth_year):
```

```
    current_year=date.today().year
```

```
    age=current_year-birth_year
```

```
    return age
```

```
return dollars
```

```
birth_year=int(input("enter birth year:"))
```

```
enter birth year:
```

```
salary_rupees=float(input("enter salary in rupees:"))
```

```
enter salary in rupees:
```

```
age=calculate_age(birth_year)
```

```
salary_dollars=convert_to_dollar(salary_rupees)
```

```
print("salary in dollars:",salary_dollars)
```



```
student={"name":"mit","roll_no":20,"percentage":90}  
print(student)  
{'name': 'mit', 'roll_no': 20, 'percentage': 90}  
student["address"]="pune"  
print(student)  
{'name': 'mit', 'roll_no': 20, 'percentage': 90, 'address': 'pune'}  
  
student.pop("percentage")  
90  
print(student)  
{'name': 'mit', 'roll_no': 20, 'address': 'pune'}  
  
student["percentage"]=95  
print(student)  
{'name': 'mit', 'roll_no': 20, 'address': 'pune', 'percentage': 95}
```

```
student={"name":"mit","roll_no":20,"percentage":90}
```

```
print(student)
```

```
{'name': 'mit', 'roll_no': 20, 'percentage': 90}
```

```
student["address"]="pune"
```

```
print(student)
```

```
{'name': 'mit', 'roll_no': 20, 'percentage': 90, 'address': 'pune'}
```

```
student.pop("percentage")
```

```
90
```

```
print(student)
```

```
{'name': 'mit', 'roll_no': 20, 'address': 'pune'}
```

```
student["percentage"]=95
```

```
print(student)
```

```
{'name': 'mit', 'roll_no': 20, 'address': 'pune', 'percentage': 95}
```